

Autobelay

long premium, short leash



An auto belay catches a climber with nobody holding the other end. Here the brake is code.

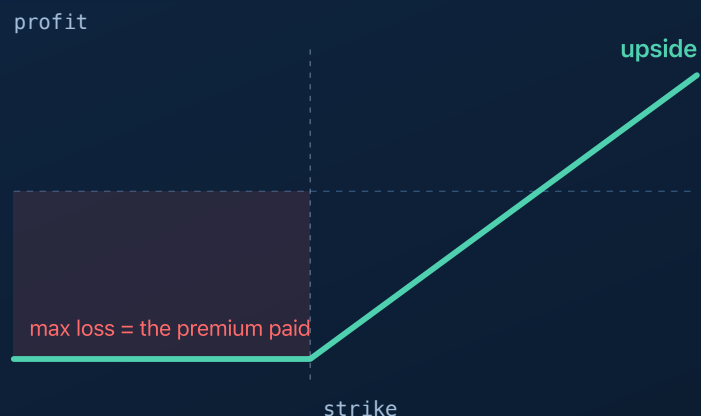
William P. McCormick / William C. McCormick · account [PA3V539Y5LE2](#) · MIT

Alpaca AI Trading Agents Hackathon · Aug 28 – Sep 4, 2026

Autobelay **long premium, short leash**

The thesis, in one sentence

Buy defined-risk, short-dated options premium when an open model can **name a reason** — and let deterministic code size it, stop it, and close it.

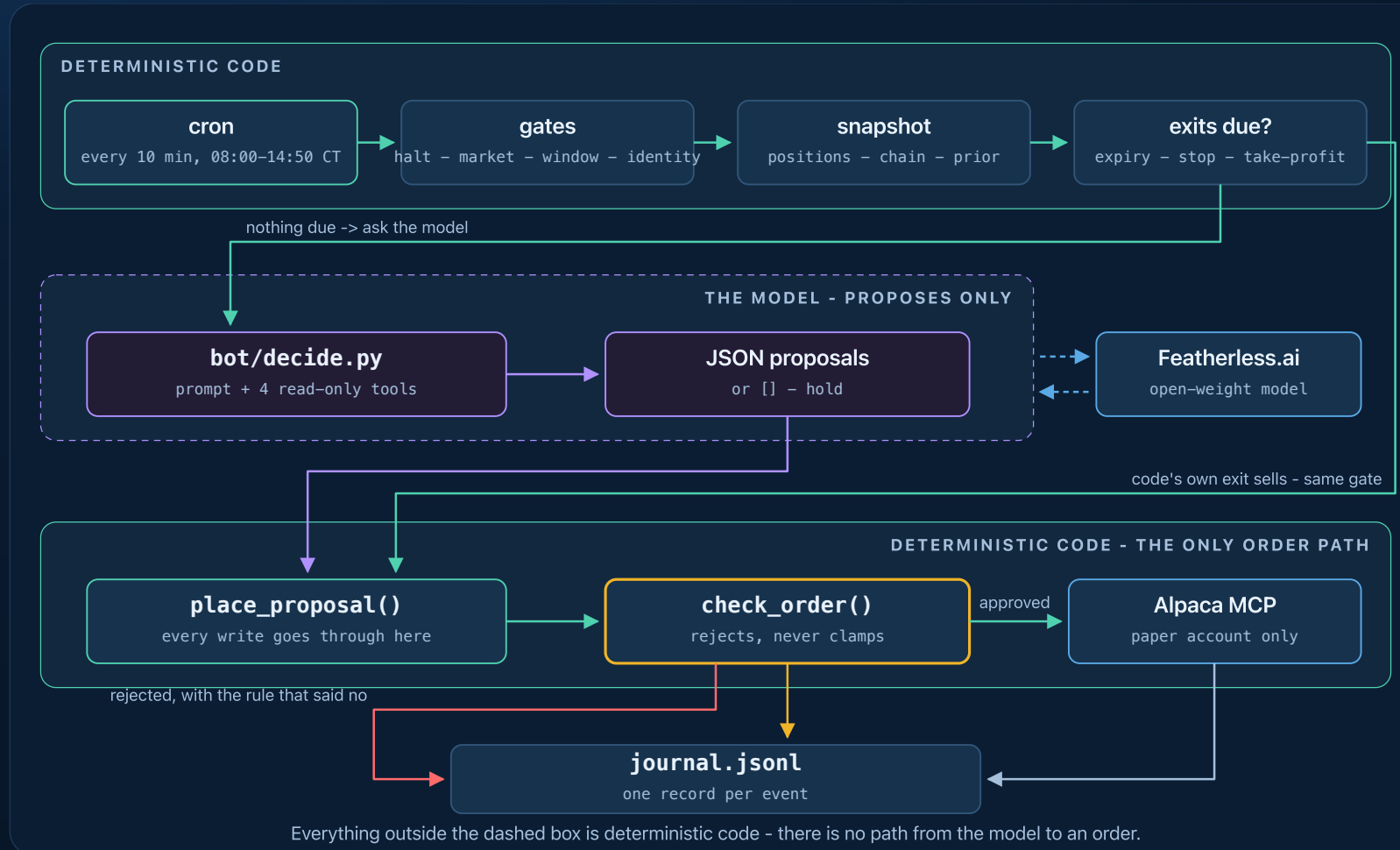


● DIVISION OF LABOUR

Long premium only, so the worst case is known before entry — which is what lets every guardrail be a flat number.

The model forms a view. It never touches an order.

One cycle, every 10 minutes



The model gets one box. Its proposals and code's own exit sells pass the

Greeks: Alpaca's, ours as the backstop

We spent three days sure the free feed carried none. It was wrong.



● FROM ALPACA

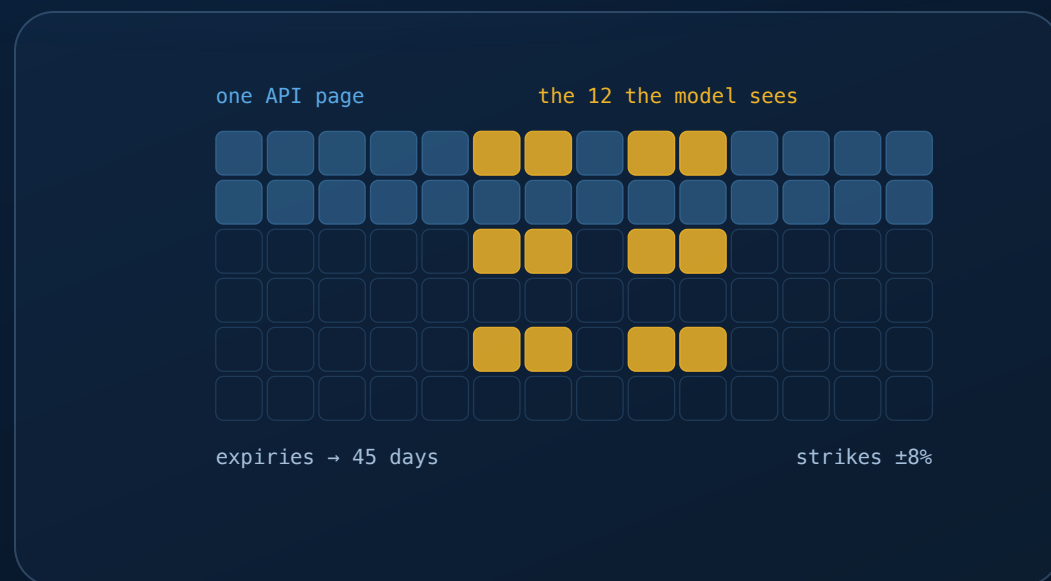
5,721 of 6,114 contracts in today's chain — one vendor surface, so a strike's put and call deltas agree.

● BLACK-SCHOLES BACKSTOP

Covers the ~6% too thin to price. Each contract carries `greeks_source`.

What the model may choose from

One API page is 500 contracts. On SPY, that is three days.



● BEFORE

500 contracts, one page, out to 3 DTE. NVDA's twelve were all one strike.

● NOW

2,912 contracts, three pages, out to 45. At-the-money *and* ~0.40 delta, three expiries, both sides — journaled every cycle.

A second opinion

Kalshi's index-close market and the chain's own implied odds. A prior to weigh, not a signal to copy — no code acts on it.

a thin market still quotes every bucket



shown
volume 756



withheld
volume 45

● KNOWING WHEN TO IGNORE IT

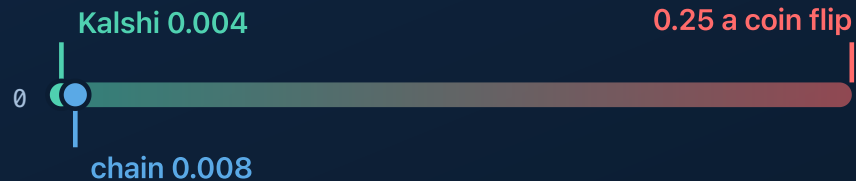
A barely-traded market quotes every bucket just as confidently, and nothing about its numbers looks broken. That is the point.

Volume and entropy gates withheld QQQ — **with the reason**, so "no second opinion today" is answerable rather than absent.

...and then we grade it

A prior nobody scores is decoration. Brier score, nightly, against what the market actually did.

Brier score - lower is better



● TUE SEP 1, JUDGED ACCOUNT

Kalshi **0.004**, the chain **0.008**. Priors the gate withheld score 0.006 — shadow-graded, so we know it discarded nothing good.

One day, and a directional one. The point is that the inputs are graded at all.

It investigates before it acts

```
research: get_bars({'symbol': 'SPY', 'timeframe': '15Min'})    -> 3334 chars
research: get_news({'symbols': 'NVDA', 'limit': 5})            -> 1770 chars
model output: [{"symbol":"NVDA260902P00220000","side":"buy","qty":10,
  "reason":"NVDA in a clear intraday downtrend (225 -> 218.6); the 220 put
    with 4 DTE, delta -0.58, aligns with the bearish momentum"}]
```

Four read-only tools, six calls at most, every one journaled. Nothing that places an order is reachable.

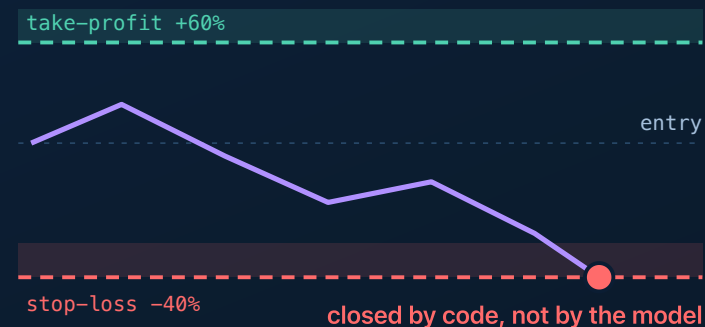
The leash

● PER PROPOSAL

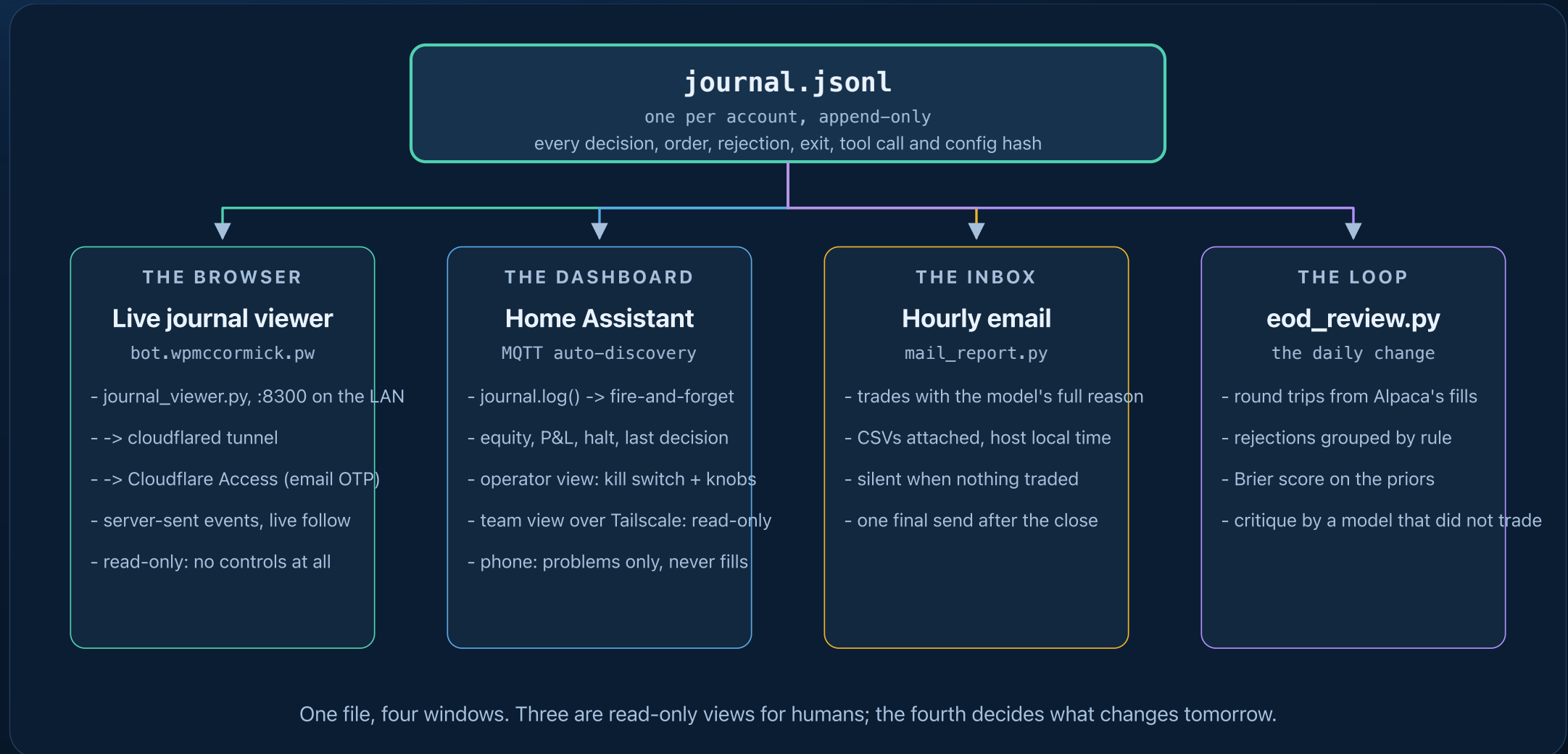
whitelist · $\leq \$5,000$ per position · ≤ 4 positions · ≤ 10 contracts
· 2–45 DTE on entries · entries 09:45–15:15 ET · a resting
buy counts as held

● PER DAY

2% loss $\rightarrow$ flatten and halt. A `HALT` file is a global kill switch.



Everything is journaled



Watching it think, from anywhere

bot.wpmccormick.pw — the journal streamed to a browser through a Cloudflare tunnel. No VPN, nothing to install.

```
14:30:17 official > ORDER  SELL 10 SPY20090606070000 @ 4.0055
Existing 8-DTE put at 760 strike is underwater (entry 4.57, current 3.92) with SPY at 761.28; thesis failed as market held above prior close despite prediction-market bearishness, and
forced close approaches with 7 DTE remaining—cut loss before theta decay accelerates and backstop forces exit.
14:30:17 official < end, 1 action(s)
14:40:09 mixed   ◆ PRIOR  SPY ref 7686.14  median 7637.5 (-0.63%) P(above) 0.069  vol 11727 → shown
◆ PRIOR  QQQ ref 29456.97 median 29050 (-1.38%) P(above) 0.058  vol 614.1 → shown
14:40:09 official ◆ PRIOR  SPY ref 7686.14  median 7637.5 (-0.63%) P(above) 0.069  vol 11727 → shown
◆ PRIOR  QQQ ref 29456.97 median 29050 (-1.38%) P(above) 0.058  vol 614.1 → shown
14:40:10 mixed   ► CYCLE  equity $100,151.92 day P&L 968.99  positions 0
14:40:10 official ► CYCLE  equity $100,209.83 day P&L 661.60  positions 0
```

It found four bugs in two days. None was a failing test — in each, the code did exactly what it had been told.

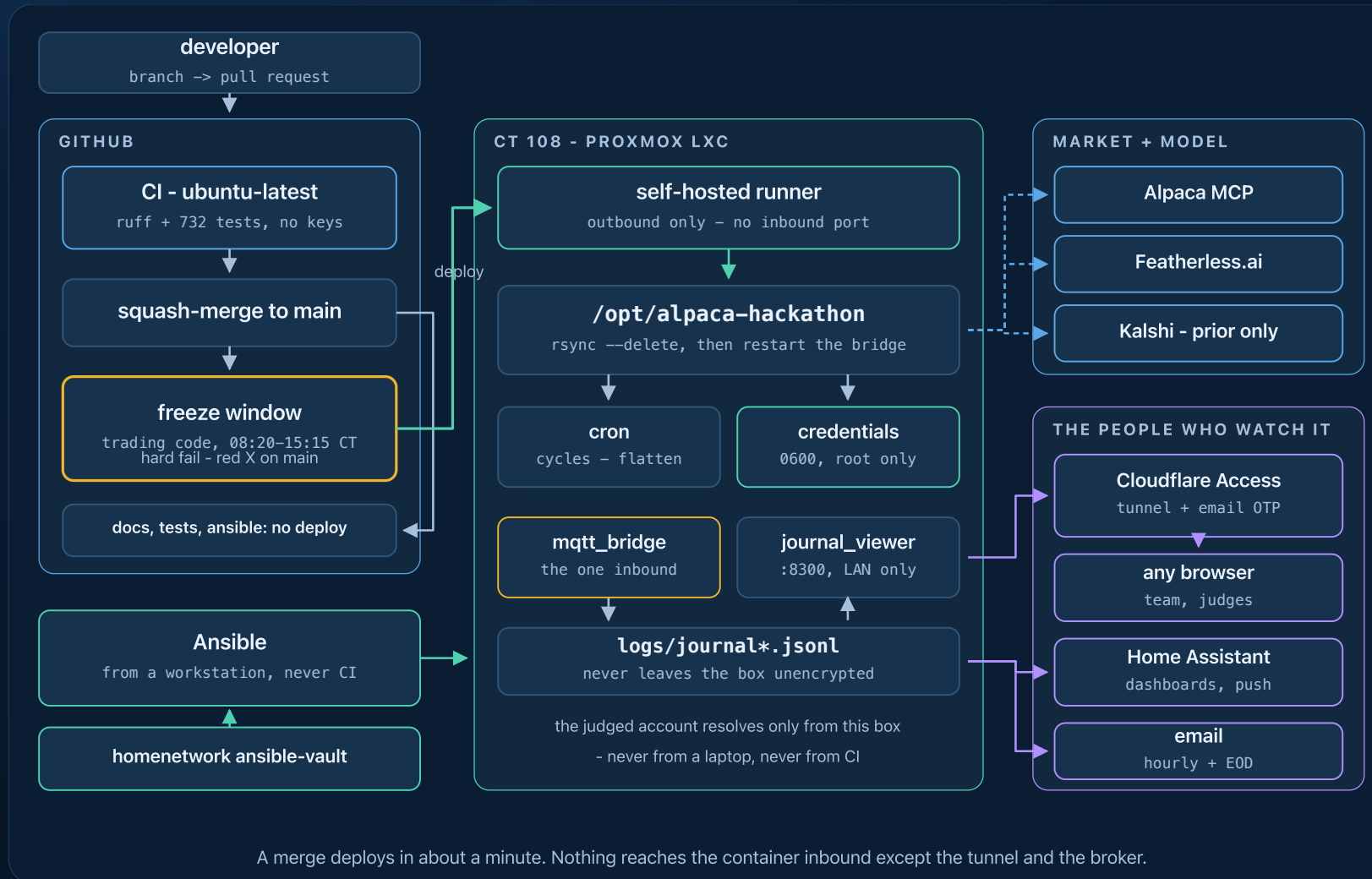
● #174 #170

The feed cut the model's reasoning mid-word.
Closing a +\$1,340 winner, it named a *neighbouring strike* — three cycles running.

● #171 #173

An unfilled limit sat invisible and the same thesis was bought again next door. A ten-minute-old position was proposed for exit — every number it cited had moved *in its favour*.

Our own hardware, not a laptop



Built to be changed while it runs

735 tests, 2s

every one runs with **no API keys and no network**

Only Docker needed

tests, a full dry-run cycle, the docs site

Rehearse any time

`--dry-run` prints orders instead of sending; `--force` skips the market gates

No deploy needed

`override.py` — allowlisted knobs, hard caps stay git-only, all of it expires at the close

Dozens of those tests exist because something broke. Each incident ends as a test.

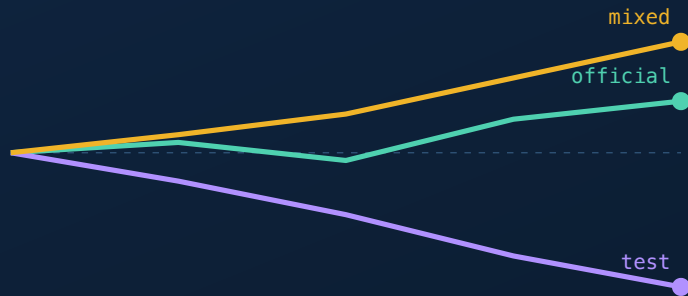
The daily loop



The critique comes from a model that did *not* trade the day. 132 PRs, 113 deploys, in seven days.

Real A/B, not a backtest

Four days is no backtest, and the free feed has no historical options data. So the experiment runs live.



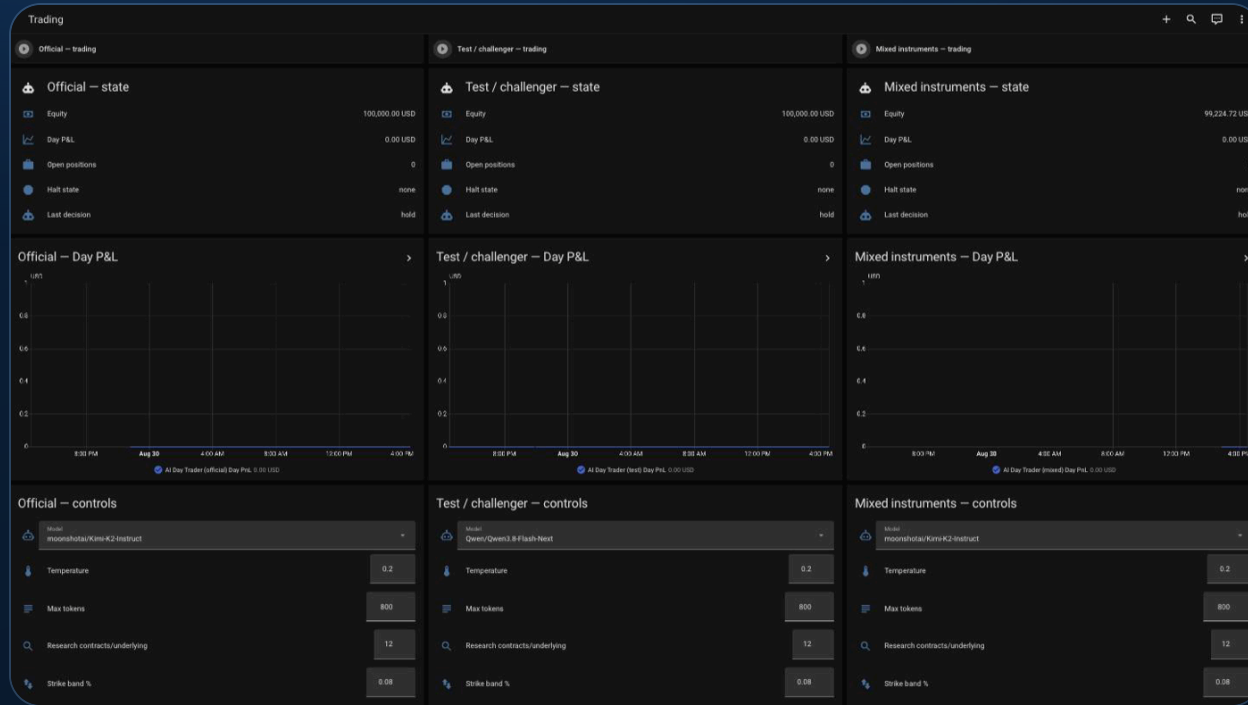
same market, same cadence, one difference each

● THREE LIVE ACCOUNTS

Same box, same ten-minute cadence, same market — one deliberate difference each. `test` swaps the model and enables research and learning; `mixed` differs by exactly one key.

The winner is promoted into the official config by pull request.

One feed, three audiences



● OPERATOR

Kill switch and knobs.

● TEAM

A read-only dashboard — Home Assistant has no per-entity permissions, so the separation has to be the dashboard.

● PHONE

Problems only. A fill sends nothing; *silence* does — no cycle for 25 minutes.

Results (fill Thu Sep 3)



Judged account Mon open → Thu close: \$____ → \$____ (____%) · ____ round trips · what didn't work: ____

What's next

- **PROMOTE WHAT THE CHALLENGERS PROVED**

Research tools, the prediction-market prior, the learning loop.

- **MULTI-LEG SPREADS**

Once the gates cover assignment risk.

github.com/bill-mccormick-dg/alpaca-hackathon · MIT

Thanks to Alpaca, lablab.ai and Featherless AI.